Proofs for the Strategic-User Model of AI Deference

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A Proofs for the Strategic-User Model

A.1 Setup and notation

The items are pieces of code that the AI has rejected in its first verdict. The code is good (G) with probability $\rho \in (0,1)$ and buggy (B) otherwise. The author observes a private signal $s \in \{g, b\}$ with $P(g \mid G) = P(b \mid B) = q \in (\frac{1}{2}, 1)$. Write $\pi_g = P(g) = \rho q + (1-\rho)(1-q)$, $\pi_b = 1 - \pi_g$ and $\mu_s = P(G \mid s)$. Bayes' rule gives

$$\pi_g \mu_g = \rho q, \qquad \pi_g(1-\mu_g) = (1-\rho)(1-q), \qquad \pi_b \mu_b = \rho(1-q), \qquad \pi_b(1-\mu_b) = (1-\rho)q, \quad (1)$$

so that $\text{odds}(\mu_g) = \text{odds}(\rho)\, q/(1-q)$ and $\text{odds}(\mu_b) = \text{odds}(\rho)\,(1-q)/q$, where $\text{odds}(x) = x/(1-x)$. Since $q > \frac{1}{2}$, $\mu_b < \rho < \mu_g$.

A final approval is worth $v_s = \beta - d\,(1-\mu_s)$ to an author with signal s, where $\beta > 0$ is the benefit of a merge and $d \geq 0$ the author's accountability for a shipped bug; a rejection is worth zero. Objecting costs $c_s \geq 0$. After an objection the AI switches to approval with probability $\delta \in [0,1]$. The author objects if and only if $\delta v_s > c_s$ (indifferent authors do not object); write $o_s(\delta)$ for this indicator and $\delta_s = c_s / v_s$ for the deference at which type s becomes willing to object.

The organization loses L_B when it approves buggy code and L_G when it rejects good code; let $K = L_B / L_G$. Switching after an objection by an author of type s reduces the expected loss by $e_s = \mu_s L_G - (1-\mu_s) L_B$. The organization's gain from a deference policy, per rejected item and relative to ignoring objections, is

$$W(\delta) = \delta \sum_{s \in \{g,b\}} \pi_s\, o_s(\delta)\, e_s. \quad (2)$$

Assumption 1.

(A1) $\text{odds}(\rho) < K$: rejecting is right on average.

(A2) $\text{odds}(\rho)\, q/(1-q) > K$: an author with a good signal is worth listening to.

(A3) $c_g v_b < c_b v_g$ whenever $v_b > 0$: a good-signal author is at least as willing to object as a bad-signal one. A3 holds whenever $c_b \geq c_g$ and $d > 0$.

Under A3, $\delta_g < \delta_b$ whenever $v_b > 0$.

A.2 Lemmas

Lemma 1 (Signs). *Under A1 and A2, $e_g > 0 > e_b$ and $\pi_g e_g + \pi_b e_b < 0$.*

Proof. For any posterior μ, $\mu L_G - (1-\mu)L_B > 0$ if and only if $\text{odds}(\mu) > K$. A2 states $\text{odds}(\mu_g) > K$, so $e_g > 0$. Since $\text{odds}(\mu_b) < \text{odds}(\rho) < K$ by A1, $e_b < 0$. Finally, $\pi_g \mu_g + \pi_b \mu_b = \rho$, so $\pi_g e_g + \pi_b e_b = \rho L_G - (1-\rho)L_B$, which is negative by A1. □

Lemma 2 (Monotone objection). *Under A3, if a bad-signal author objects at some δ, so does a good-signal author.*

Proof. If $\delta v_b > c_b$ with $v_b > 0$, then $\delta > \delta_b > \delta_g$ by A3, hence $\delta v_g > c_g$ (note $v_g \geq v_b > 0$ because $\mu_g > \mu_b$). □

Lemma 3 (First-best bound). *Under A1 and A2, for every $\delta \in [0,1]$ and every pattern of objections, $W(\delta) \leq \pi_g e_g$, with equality only if $\delta = 1$ and only good-signal authors object.*

Proof. $W(\delta) = \delta\,[\pi_g o_g e_g + \pi_b o_b e_b] \leq \delta\,\pi_g e_g \leq \pi_g e_g$, using $e_b < 0 < e_g$ (Lemma 1). Both inequalities hold with equality only if $o_b = 0$, $o_g = 1$ and $\delta = 1$. □

A.3 Propositions under commitment

Throughout this subsection the designer fixes δ (in training) before authors decide.

Proposition 1 (Cheap talk unravels). *Under A1 and A2, if $c_b = 0$ and $v_b > 0$, then $W(\delta) < 0$ for every $\delta \in (0,1]$, so the optimal deference is $\delta^* = 0$.*

Proof. For $\delta > 0$, $\delta v_b > 0 = c_b$, so $o_b = 1$. If $o_g = 1$, $W(\delta) = \delta(\pi_g e_g + \pi_b e_b) < 0$ by Lemma 1; if $o_g = 0$, $W(\delta) = \delta \pi_b e_b < 0$. Since $W(0) = 0$, the unique maximizer is $\delta = 0$. □

Proposition 2 (Optimal deference). *Under A1–A3, if $v_b > c_b > 0$, then W is uniquely maximized at*

$$\delta^* = \frac{c_b}{v_b} < 1, \qquad W(\delta^*) = \delta^* \pi_g e_g > 0.$$

Proof. Since $v_g \geq v_b > 0$, the thresholds satisfy $\delta_g < \delta_b = \delta^* < 1$ and split $[0,1]$ into three intervals. On $[0, \delta_g]$, $\delta v_s \leq c_s$ for both types, so nobody objects and $W = 0$. On $(\delta_g, \delta_b]$, only good-signal authors object, so $W(\delta) = \delta \pi_g e_g$, which is positive and strictly increasing by Lemma 1. On $(\delta_b, 1]$, bad-signal authors object and, by Lemma 2, so do good-signal authors; hence $W(\delta) = \delta(\pi_g e_g + \pi_b e_b) < 0$. The maximum is attained at δ_b and nowhere else. □

If indifferent authors objected instead, the supremum would not be attained, and any δ slightly below δ^* would come arbitrarily close to it.

Proposition 3 (The cliff). *Under the conditions of Proposition 2, $W(\delta) < 0$ for every $\delta > \delta^*$, and*

$$W(\delta^*) - \lim_{\delta \downarrow \delta^*} W(\delta) = \delta^* \pi_b \big[(1-\mu_b)L_B - \mu_b L_G\big] > 0.$$

Proof. Both claims follow from the third interval in the proof of Proposition 2: just above δ^*, $W(\delta) = \delta(\pi_g e_g + \pi_b e_b)$, and the jump equals $-\delta^* \pi_b e_b$, which is positive by Lemma 1. Beyond δ^* the loss grows linearly in δ. □

Proposition 4 (Comparative statics)**.** *On the region $v_b > c_b > 0$, the optimal deference $\delta^* = c_b/[\beta - d(1-\mu_b)]$ and the optimal gain $W^* = \delta^* \pi_g e_g$ satisfy*

$$\frac{\partial \delta^*}{\partial \beta} = -\frac{c_b}{v_b^2} < 0, \qquad \frac{\partial \delta^*}{\partial d} = \frac{c_b(1-\mu_b)}{v_b^2} > 0, \qquad \frac{\partial \delta^*}{\partial c_b} = \frac{1}{v_b} > 0,$$

and W^* moves in the same direction, because $\pi_g e_g$ does not depend on β, d or c_b. Both δ^* and W^* also increase with the author's signal precision q.

Proof. The derivatives follow directly from $\delta^* = c_b/v_b$. For q: $\text{odds}(\mu_b) = \text{odds}(\rho)(1-q)/q$ falls in q, so v_b falls and δ^* rises; and by (1), $\pi_g e_g = \rho q L_G - (1-\rho)(1-q)L_B$ rises in q. □

Proposition 5 (Raising the cost of groundless objections)**.** *Under A1–A3 and $v_g > c_g$: (a) δ^* and W^* are non-decreasing in c_b, and strictly increasing while $c_b < v_b$; (b) if $c_b \geq v_b$, then $\delta^* = 1$ and $W(1) = \pi_g e_g$, the first best of Lemma 3.*

Proof. (a) For $c_b < v_b$ this is Proposition 4. (b) If $c_b \geq v_b$, then $\delta v_b \leq v_b \leq c_b$ for every $\delta \leq 1$, so bad-signal authors never object, and $W(\delta) = \delta \pi_g e_g$ for $\delta > \delta_g$. This is maximized at $\delta = 1$, where it equals the bound of Lemma 3. As c_b rises to v_b, $\delta^* = c_b/v_b$ rises continuously to 1, so W^* is non-decreasing throughout. □

Corollary 1 (An evidence window)**.** *Suppose an objection must be backed by evidence that is free to produce for good code and costs $F \geq 0$ for buggy code, so that $c_s = c + F(1-\mu_s)$ with $c \geq 0$. Assume $\beta > c$, $d > 0$ and $c + F > 0$, and define*

$$\underline{F} = \frac{\beta - c}{1-\mu_b} - d, \qquad \overline{F} = \frac{\beta - c}{1-\mu_g} - d, \qquad \underline{F} < \overline{F}.$$

Then (i) for $F < \underline{F}$, $\delta^* = c_b/v_b < 1$ and rises with F; (ii) for $\underline{F} \leq F < \overline{F}$, $\delta^* = 1$ and pushback attains the first best; (iii) for $F \geq \overline{F}$, nobody objects at any δ and pushback has no value.

Proof. Here $c_b - c_g = F(\mu_g - \mu_b) \geq 0$ and $d > 0$, so A3 holds. The condition $c_b \geq v_b$ is equivalent to $F \geq \underline{F}$, and $v_g > c_g$ is equivalent to $F < \overline{F}$; the ordering $\underline{F} < \overline{F}$ follows from $\beta > c$ and $1 - \mu_g < 1 - \mu_b$. Case (i) is Proposition 2 (with $c_b > 0$ because $c + F > 0$), and case (ii) is Proposition 5(b). In case (iii), $v_g \leq c_g$ and $v_b < v_g \leq c_g \leq c_b$, so neither type objects for any $\delta \leq 1$ and $W \equiv 0$. □

A.4 Equilibria without commitment

Without commitment, the AI chooses δ after observing an objection to minimize the organization's expected loss given its posterior, and authors choose objection probabilities $z_s \in [0,1]$. We study perfect Bayesian equilibria and assume $c_g, c_b > 0$.

Proposition 6 (Value of commitment)**.** *Under A1–A3 and $v_b > c_b > 0$:*

1. *There is an equilibrium in which nobody objects and $\delta = 0$, supported by any off-path belief whose odds do not exceed K.*

2. *In every equilibrium in which objections occur with positive probability, $z_g = 1$, $\delta = c_b/v_b$, and*

$$z_b = z^* = \frac{\rho q - K(1-\rho)(1-q)}{K(1-\rho)q - \rho(1-q)} \in (0,1).$$

3. In every equilibrium, $W = 0$. The value of commitment is therefore $W(\delta^*) = \delta^* \pi_g e_g > 0$.

Proof. Part 1. With $\delta = 0$ every author's payoff from objecting is $-c_s < 0$, so nobody objects; a posterior with odds at most K makes $\delta = 0$ optimal after the off-path objection.

Part 2. Let O be the posterior odds that the code is good after an objection. If $O > K$, the AI sets $\delta = 1$. Then $\delta v_b = v_b > c_b$, so $z_b = 1$, and $z_g = 1$ by Lemma 2; by the Bayes identities (1), $O = \text{odds}(\rho) < K$, a contradiction. If $O < K$, the AI sets $\delta = 0$ and nobody objects, contradicting that objections occur. Hence $O = K$ and the AI is indifferent over δ.

Next, $z_b \in (0, 1)$. If $z_b = 0$, objections come only from good-signal authors, so $O = \text{odds}(\mu_g) > K$ by A2. If $z_b = 1$, then $\delta v_b \geq c_b$, good-signal authors strictly prefer to object by A3, and $O = \text{odds}(\rho) < K$. Both contradict $O = K$. A bad-signal author mixes only if indifferent, so $\delta = c_b / v_b$; at this δ, good-signal authors strictly prefer to object by A3, so $z_g = 1$. By the Bayes identities (1),

$$O = \frac{\rho q + z\rho(1-q)}{(1-\rho)(1-q) + z(1-\rho)q} = K$$

gives $z = z^*$. The numerator of z^* is positive by A2 and the denominator is positive because $\text{odds}(\mu_b) < K$; $z^* < 1$ rearranges to $\rho < K(1-\rho)$, which is A1.

Part 3. The gain is $W = \delta \sum_s \pi_s z_s e_s = \delta \left[L_G \, P(G, \text{objection}) - L_B \, P(B, \text{objection}) \right]$, which is zero when $O = K$, and zero trivially when nobody objects. Comparing with Proposition 2 gives the value of commitment. □

Remark 1. Commitment matters only in the interior case. If $c_b \geq v_b$, bad-signal authors never object, the posterior after an objection is μ_g, and $\delta = 1$ is sequentially rational, so the first best is reached without commitment. Training thus has its largest leverage exactly when authors are biased enough that groundless objections are tempting.

All propositions, the first-best bound and the evidence window were also verified numerically on random parameter draws satisfying A1–A3 (`strategic_user.py --check`).